

Deva Anand

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Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Performance Engineering, Systems for Data Science, Optimization in CS, Advanced Machine Learning, Advanced NLP

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, Rust, C++, JavaScript, TypeScript, Java, Go, SQL, Bash

Backend & Services

Node.js, Express, FastAPI, Fastify, REST APIs, async services, integration testing

Tooling & Performance

Spark / PySpark, Spark UI profiling, performance tuning, pytest, Git, GitHub Actions, CI/CD

Cloud & DevOps

AWS, GCP, Azure, Docker, Linux

Databases & Web

MongoDB, PostgreSQL, Redis, SQLite, React

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Designed and built a **full-stack internal tool** (React, Node.js, MongoDB) used daily by the AI team to label data for production models, replacing error-prone spreadsheet workflows with secure role-based REST APIs; cut manual operations by **100+ hours/month** and improved throughput by **60%**.
- Engineered a **multi-tenant backend** (Node.js, MongoDB) for **8+ clients** processing **10,000+ monthly submissions**, designed for maintainability and scale and integrated with an external platform via REST APIs; owned reliability and remediated **20+ VAPT findings**.
- Served as primary technical point of contact, partnering across teams on releases and integrations, debugging production issues for performance and reliability, and running knowledge transfer for new engineers.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and shipped **20+ RESTful APIs** in Node.js and Fastify over **PostgreSQL** with input validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated the core codebase from **JavaScript to TypeScript** across multiple repositories for type safety and long-term maintainability, and integrated **LLM-powered** context-aware replies into a chat product, iterating on prompt design and response-quality heuristics.

Projects

AgenticSearch - Provenance-First Entity Discovery Service

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a Python/FastAPI service running a multi-stage pipeline (typed retrieval planning, deterministic parsers with LLM fallback, output verification, cell-level provenance) that turns open-ended queries into structured, auditable output.
- Hardened the tool with **150+ automated tests**, provider routing and fallback, and pipeline instrumentation for reliability and debuggability.

Spark ETL Optimization - Large-Scale Data Pipeline

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI | 1.4B-row NYC TLC dataset

- Cut end-to-end ETL runtime **25%** (17–19 min to 14.3 min) on a **1.4B-row** dataset via broadcast joins (minutes to 0.05s), AQE, skew-join handling, shuffle-partition tuning, schema-aware reads, and year/month partitioning.
- Used **Spark UI** stage timings to debug shuffle and skew bottlenecks for performance and reliability, cutting tail-latency ratio (P99/P50) from **2.14x to 1.74x** and documenting a repeatable optimization playbook.
- Added a 6-rule **data-quality filter** (retaining **95.26%** of records) and a parallel data downloader, keeping the dataset clean and the pipeline reproducible across naive and tuned benchmark runs.

Sonare - Offline Sign ↔ Speech Cross-Platform Desktop App

github.com/Deva-1903/Qualcomm-Sep25-Team-Sonare

React, Electron, FastAPI, MediaPipe, whisper.cpp | Qualcomm Edge AI Hackathon 2025

- Built a **full-stack desktop application** end-to-end, integrating camera, microphone, UI, and backend into an on-device pipeline (MediaPipe hand tracking, whisper.cpp ASR, local Text-To-Speech), driven from concept to a shipped product under hackathon time pressure.
- Engineered async APIs and queues to keep the UI responsive while streaming local video and audio with no external network dependencies.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.